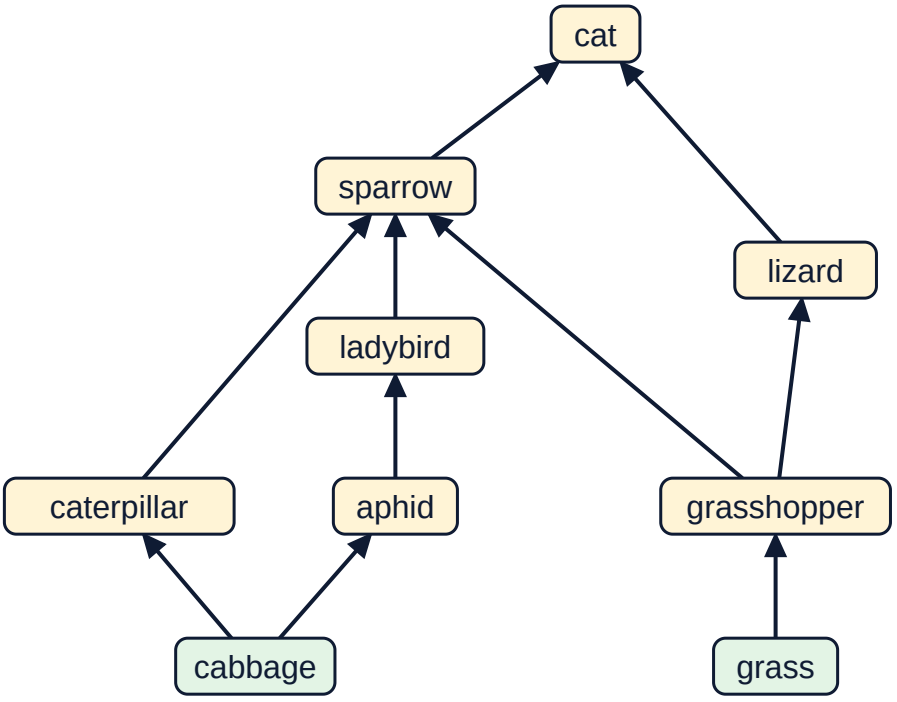
Checkpoint 6 Science — Practice Test 1 — Mark Scheme

Paper 1: 35 minutes · calculator allowed · Paper 2: 35 minutes · Total 80 marks

Abdelkriem Academy · practice paper written in the style of Cambridge Checkpoint. Not an official Cambridge paper.

Paper 1 — 35 minutes — 40 marks

1 Pupils find these living things in the school garden. The food web shows what eats what. **[1]**



(a) What is the source of energy for this food web?

Answer: Sun or the Sun or sunlight or light or light energy or the sun's light or light from the Sun or energy from the Sun or solar energy or the sun's energy
Notes / working: The **Sun**. The producers (cabbage and grass) trap light energy from the Sun to make their food; every animal in the web gets its energy from them.

(b) How many food chains in this food web include the sparrow? **[1]**

Answer: 3
Notes / working: Follow every path that passes through the sparrow:
1. cabbage → caterpillar → sparrow → cat
2. cabbage → aphid → ladybird → sparrow → cat
3. grass → grasshopper → sparrow → cat
So **3** food chains. (The fourth chain, grass → grasshopper → lizard → cat, does not include the sparrow.)
Tip: always write the chains out — it is easy to miss one.

(c) Which food chain is in this food web? **[1]**

- ☐ A) cabbage → grasshopper → lizard → cat
- ☐ B) grass → grasshopper → lizard → cat
- ☐ C) aphid → cabbage → ladybird → sparrow
- ☐ D) grass → caterpillar → sparrow → cat

Answer: B) grass → grasshopper → lizard → cat
Notes / working: Check each arrow in the web. Grass → grasshopper → lizard → cat is correct (B). A is wrong: the grasshopper eats grass, not cabbage. C starts with an animal and has the arrow the wrong way (the aphid eats the cabbage). D is wrong: the caterpillar eats cabbage, not grass.

(d) The gardener sprays the cabbages with a pesticide to kill the aphids. The pesticide is toxic and it stays in animals' bodies. **[2]**
Explain why the cat could be harmed, even though it never eats cabbage.

Answer: 1 mark: the pesticide/toxin passes along the food chain, e.g. aphid → ladybird → sparrow → cat (accept: animals that ate the cabbage/aphids are eaten by animals higher up the chain). 1 mark: each animal eats many of the animals below it, so the amount of toxin builds up / the cat collects a large amount of toxin. Do not accept: 'the cat eats the pesticide' with no food chain idea.
Notes / working: The aphids (and caterpillars) take in the pesticide when they eat the cabbage. Ladybirds eat many aphids, sparrows eat many ladybirds and caterpillars, and the cat eats sparrows. The toxin **moves along the food chain** and, because each animal eats many of the ones below it, the amount **builds up** — so the cat can get a harmful amount without ever touching a cabbage.

2 The garden club wants to know if adding compost to sandy soil helps bean plants grow taller. They fill pots with sandy soil and mix in different amounts of compost. They plant one bean seed in each pot and use three pots for each amount of compost. After four weeks they measure the height of each plant. **[1]**

Compost added (spoonfuls)	Pot 1 height (cm)	Pot 2 height (cm)	Pot 3 height (cm)
0	12	13	11
2	17	18	16
4	21	9	22
6	24	25	23

(a) What is the independent variable in this investigation? (the variable the club changes)

- ☐ A) the height of the plants
- ☐ B) the type of soil
- ☐ C) the amount of compost
- ☐ D) the number of pots

Answer: C) the amount of compost
Notes / working: The club **changes** the amount of compost (0, 2, 4, 6 spoonfuls), so that is the independent variable. The height is what they measure (dependent variable). The type of soil is kept the same (sandy) — a control variable.

(b) Give **one** variable the club must keep the same to make this a fair test. **[1]**

Answer: 1 mark for any one: type of soil / amount (volume or mass) of soil / size of pot / type of bean seed / amount of water / how often they water / amount of light / place or temperature / time the plants grow (four weeks). Do not accept: amount of compost (this is changed); height of plant (this is measured); 'the same pots' with no property.
Notes / working: Any variable that could also affect plant growth must be kept the same, for example **the amount of water each pot gets**, the size of the pots, the type of bean, or where the pots are kept (light and temperature).

(c) One result is anomalous (it does not fit the pattern). Write down this height. **[1]**

Answer: 9 cm
Notes / working: With 4 spoonfuls, pots 1 and 3 grew 21 cm and 22 cm, but pot 2 grew only **9 cm** — much less than the other two pots, and even less than the plants with no compost. It does not fit the pattern, so it is anomalous (perhaps that seed was damaged or the pot was not watered).

(d) Calculate the mean height of the plants with 2 spoonfuls of compost. **[1]**

Answer: 17 cm
Notes / working: Mean = total ÷ number of results = (17 + 18 + 16) ÷ 3 = 51 ÷ 3 = **17 cm**.

(e) Write a conclusion for this investigation. **[1]**

> **Answer:** 1 mark: the more compost (added to the sandy soil), the taller the bean plants grew / ora. Accept: adding compost helps the bean plants grow taller. Do not accept: 'compost is good' or a statement about one pot only.
> **Notes / working: The more compost added to the sandy soil, the taller the bean plants grew** (mean heights about 12 cm → 17 cm → 21.5 cm → 24 cm, leaving out the anomalous 9 cm).

(f) Use your science to explain why adding compost helps plants to grow in sandy soil. **[1]**

> **Answer:** 1 mark for one of: compost is organic matter that adds nutrients/minerals to the soil; OR compost holds water so the sandy soil does not dry out as quickly (sandy soil drains quickly). Do not accept: 'compost is food for plants' without nutrients/minerals.
> **Notes / working:** Sandy soil has large grains and big gaps, so water drains through quickly and it has few nutrients. Compost is **organic matter** (rotted plants): it adds **nutrients (minerals)** and **holds water**, so the soil can support more plant growth.

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3 Pupils take soil from three parts of the school garden and find out what each soil contains. **[1]**

Soil	Sand (%)	Clay (%)	Organic matter (%)
X	80	10	10
Y	15	75	10
Z	40	25	35

(a) Which soil is a clay soil? Write X, Y or Z.

> **Answer:** Y or soil Y
> **Notes / working:** Soil **Y** has 75% clay — far more clay than the others, so it is a clay soil.

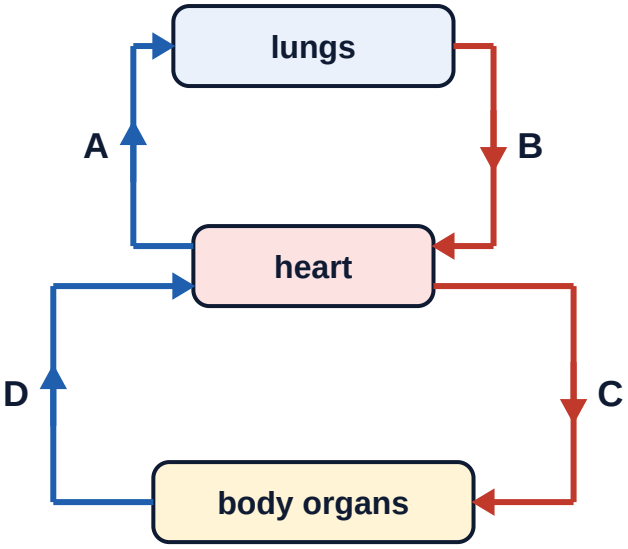
(b) Water drains through one soil the fastest. Which soil? Write X, Y or Z. **[1]**

> **Answer:** X or soil X
> **Notes / working:** Soil **X** is 80% sand. Sand grains are large with big gaps between them, so water drains through sandy soil fastest. Clay particles are the smallest; a soil with lots of clay holds water and drains slowly.

(c) Soil X dries out very quickly in summer. Suggest what the gardener could add to soil X to help it hold more water. **[1]**

> **Answer:** 1 mark: compost / organic matter / humus / leaf mould / animal manure. Accept: clay. Do not accept: water / sand / fertiliser.
> **Notes / working:** Add **compost (organic matter)**. It soaks up and holds water like a sponge, so the sandy soil does not dry out so quickly (and it adds nutrients too).

4 The diagram shows how blood flows around the human body. The arrows show the direction of blood flow. **[1]**



(a) What type of blood vessel is D?

- ☐ A) artery
- ☐ B) capillary
- ☐ C) trachea
- ☐ D) vein

Answer: D) vein
Notes / working: Vessel D carries blood from the body organs **back to the heart**. Vessels that carry blood back to the heart are **veins** (D). Arteries carry blood away from the heart; capillaries are the tiny vessels inside the organs; the trachea is the windpipe, not a blood vessel.

(b) Which vessel, A or B, carries blood that has just collected oxygen? **[1]**

Answer: B or vessel B
Notes / working: Blood collects oxygen in the **lungs**. Vessel **B** carries blood from the lungs back to the heart, so its blood is full of oxygen. Vessel A carries blood to the lungs, before it has collected oxygen.

(c) Name the tiny blood vessels inside the body organs, where oxygen and nutrients pass from the blood into the cells. **[1]**

Answer: capillaries or capillary or capilaries or capillarys
Notes / working: Capillaries. Their walls are very thin, so oxygen and nutrients can pass out to the cells, and waste can pass in.

(d) Name the waste gas that the blood carries from the body cells to the lungs. **[1]**

> **Answer:** carbon dioxide or CO2 or CO_2 or carbon-dioxide or carbondioxide
>
> **Notes / working: Carbon dioxide**. It is made by the cells, carried by the blood to the lungs and breathed out.

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5 The school garden has a barrel of rainwater. In summer, mosquitoes lay their eggs in the still water. Some kinds of mosquito can carry the parasite that causes malaria. **[2]**

(a) Describe **two** ways the pupils can avoid being bitten by mosquitoes in the garden.

> **Answer:** 1 mark each for any two different ways: put a lid on the barrel / cover or empty still water (so mosquitoes cannot breed); use insect repellent (on skin); wear long sleeves and long trousers / cover the skin; use mosquito nets or screens; avoid the garden at dawn/dusk when mosquitoes bite most. Do not accept: 'wash hands' / 'take medicine' / two answers that are the same idea.
> **Notes / working:** For example: **(1) put a lid on the water barrel (or empty it) so mosquitoes cannot lay eggs there**; **(2) wear long sleeves and trousers or put on insect repellent**. Other good answers: use nets or screens, avoid the garden at dusk.

(b) The pupils pick tomatoes from the garden to eat. Explain why they should wash their hands and the tomatoes first. **[1]**

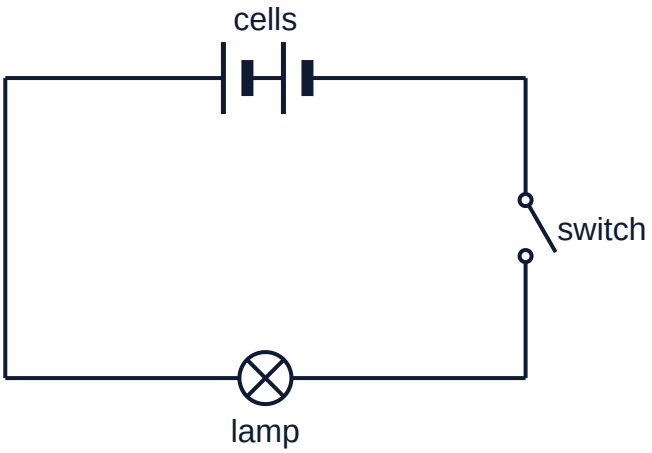
> **Answer:** 1 mark: (soil/dirt on hands or tomatoes may carry) bacteria/germs/pathogens/microorganisms/parasites AND washing removes them so they are not swallowed / so they do not cause disease. Do not accept: 'to remove dirt' alone.
> **Notes / working:** Soil and dirty hands can carry **pathogens (germs such as bacteria or parasite eggs)**. Washing removes them, so they are not swallowed with the food and cannot cause disease.

(c) Mariam scratches her arm on a rose thorn and it bleeds. Which of the body's defences against pathogens has been broken? **[1]**

- ☐ A) skin
- ☐ B) mucus
- ☐ C) stomach acid
- ☐ D) saliva

> **Answer:** A) skin
> **Notes / working:** The **skin** is a barrier that stops pathogens getting into the body. A cut makes a gap in it, so the cut should be cleaned and covered. Mucus traps pathogens in the nose and airways; stomach acid kills pathogens in swallowed food.

6 Youssef takes apart his toy torch. Inside there are two cells, a switch and a lamp. He draws this circuit diagram of the torch. **[1]**



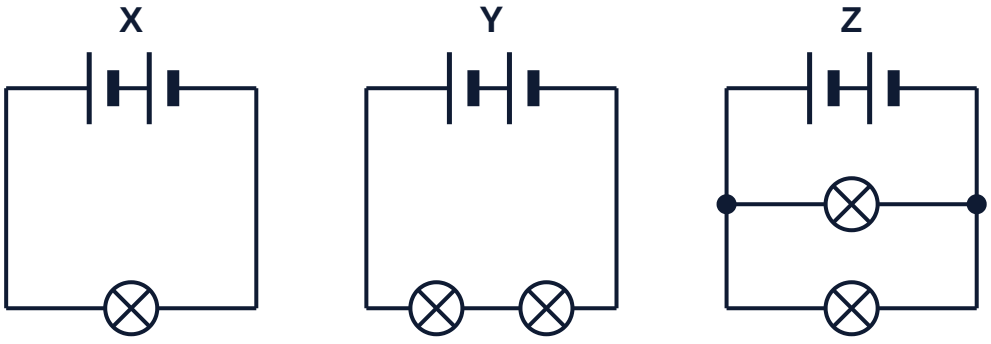
(a) Is the switch in Youssef's diagram open or closed?

Answer: open or it is open or open switch
Notes / working: The switch symbol has a gap — the lever is lifted up and does not touch the second contact — so the switch is **open**.

(b) Explain why the lamp in Youssef's diagram is not lit. **[1]**

Answer: 1 mark: the switch is open so the circuit is not complete / there is a gap (break) in the circuit, so electricity (current) cannot flow. Do not accept: 'the cells are flat' / 'the lamp is broken'.
Notes / working: The switch is open, so there is a **gap in the circuit**. The circuit is not complete, so no current can flow through the lamp.

7 (continued) **[1]**
Youssef builds three circuits, X, Y and Z. All the cells and lamps are the same as in the torch.



(c) In which circuit, X, Y or Z, are the lamps the dimmest?

Answer: Y or circuit Y
Notes / working: In **Y** the two lamps are in **series**: the same cells must push the current through both lamps one after the other, so each lamp is dimmer than the single lamp in X. In Z the lamps are in parallel, so each one is as bright as the lamp in X.

(d) Youssef wants a torch with two lamps that are both as bright as the lamp in circuit X. Which circuit should he copy, Y or Z? **[1]**

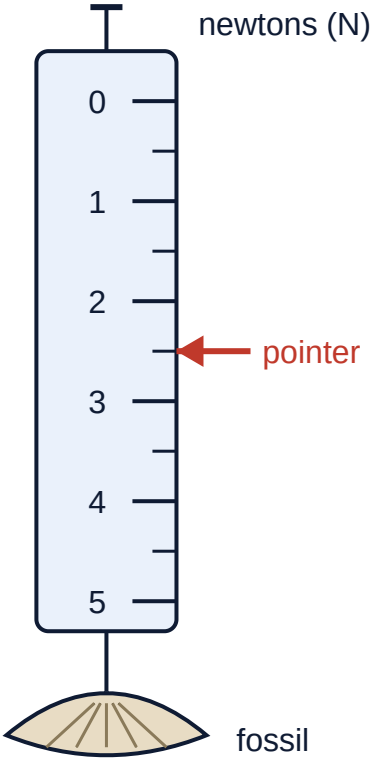
> **Answer:** Z or circuit Z
> **Notes / working:** Circuit **Z** (parallel). Each lamp in a parallel circuit is as bright as one lamp on its own.

(e) Explain why the lamps in circuit Z are brighter than the lamps in circuit Y. **[1]**

> **Answer:** 1 mark: in Z (parallel) each lamp is connected directly to the cells / has its own path (loop) to the cells / gets the full push of the cells, but in Y (series) the lamps share the push of the cells / the current must go through both lamps. Accept: 'in Y the energy is shared between two lamps'.
> **Notes / working:** In the parallel circuit Z, **each lamp has its own path to the cells**, so each lamp gets the full push of the two cells. In the series circuit Y, the two lamps **share** the push of the cells, so each one is dimmer.

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8 On a fossil hunt in the Western Desert, Hana finds a fossil sea urchin. At school she hangs it from a forcemeter. **[1]**



(a) What is the weight of the fossil? Read the forcemeter.

> **Answer:** 2.5 N
> **Notes / working:** Each numbered mark is 1 N and there is one small mark halfway between, so each small division is 0.5 N. The pointer is halfway between 2 and 3, so the weight is **2.5 N**.

(b) Gravity on Earth is 10 N/kg. Calculate the mass of the fossil in **grams**. **[1]**

> **Answer:** 250 g
> **Notes / working:** Mass = weight ÷ gravity = 2.5 ÷ 10 = 0.25 kg.
> 0.25 kg × 1000 = **250 g**.
> Common mistake: giving 0.25 — the question asks for grams. Another mistake: 2.5 × 10 = 25 (multiplying instead of dividing).

(c) A space robot takes the same fossil to the Moon, where gravity is weaker. Which statement is correct? **[1]**

- ☐ A) Its mass and its weight both stay the same.
- ☐ B) Its mass and its weight are both less.
- ☐ C) Its mass stays the same but its weight is less.
- ☐ D) Its mass is less but its weight stays the same.

> **Answer:** C) Its mass stays the same but its weight is less.
> **Notes / working: Mass** is the amount of matter in the fossil — that does not change when it moves. **Weight** is the pull of gravity on it, so weaker gravity on the Moon means a smaller weight. Answer C.

(d) Gravity on the Moon is 1.6 N/kg. Calculate the weight of the fossil on the Moon. **[1]**

> **Answer:** 0.4 N
> **Notes / working:** Weight = mass (in kg) × gravity = 0.25 kg × 1.6 N/kg = **0.4 N**.
> Common mistake: using 250 (grams) instead of 0.25 kg, giving 400.

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9 Hana's group also finds three rocks, J, K and L, in the Western Desert. **[1]**

Rock	What it looks like
J	small grains and tiny pieces of shell stuck together in layers; a shell fossil; crumbles when rubbed
K	interlocking crystals of different colours; no layers; very hard
L	crystals lined up in wavy bands; no fossils; very hard

(a) Which rock is igneous? Write J, K or L.

> **Answer:** K or rock K
> **Notes / working: K**. Igneous rock forms when melted rock (magma or lava) cools and solidifies, so it is made of **interlocking crystals** in no particular pattern, with no layers and no fossils.

(b) What type of rock is L? **[1]**

> **Answer:** metamorphic or metamorphic rock or metamorphic rocks
> **Notes / working: Metamorphic**. Heat and pressure deep underground change a rock without melting it; the crystals line up in bands, and any fossils are destroyed.

(c) Rock J is a sedimentary rock. Describe how the shell fossil in rock J was formed. **[2]**

Answer: 1 mark: a sea animal died and its shell sank to the sea floor and was (quickly) covered by sediment (sand/mud). 1 mark: more layers of sediment built up on top and squeezed/pressed the layers into rock; the shell was replaced by minerals / turned to stone (accept: left its shape/print in the rock). Marks are independent.
Notes / working: 1. A sea animal died and its shell sank to the **sea floor**, where it was quickly **buried by sediment** (sand and mud).
2. Over a very long time, **more layers** of sediment built up on top. Their weight **squeezed the layers into rock**, and **minerals** slowly replaced the shell, so it became a fossil in stone.

(d) Hana wants a quick test to help tell rock J from rock K. Which test is best? **[1]**

- ☐ A) Measure the temperature of each rock.
- ☐ B) Weigh each rock on a balance.
- ☐ C) Look at the colour of each rock.
- ☐ D) Drip water onto each rock and see if it soaks in.

Answer: D) Drip water onto each rock and see if it soaks in.
Notes / working: Sedimentary rock J has small gaps between its grains, so water **soaks in**. Igneous rock K has tightly interlocking crystals, so water stays on the surface. Temperature, mass and colour do not depend on the rock type, so they cannot tell the rocks apart. Answer D.

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10 At school, Hana drips vinegar onto a piece of rock J. The vinegar fizzes and bubbles of carbon dioxide gas are given off. **[1]**
Rock J + vinegar → new substances, including carbon dioxide

(a) What are the reactants in this reaction?

- ☐ A) carbon dioxide only
- ☐ B) rock J and vinegar
- ☐ C) bubbles and water
- ☐ D) rock J only

Answer: B) rock J and vinegar
Notes / working: Reactants are the substances you start with, written on the left of the arrow: rock J (the limestone in it) and vinegar. Carbon dioxide is a **product** — a new substance that is made.

(b) Give the evidence that shows a chemical reaction happened. **[1]**

> **Answer:** fizzing or it fizzes or bubbles or bubbles of gas or a gas is produced or gas produced or gas or a gas is given off or gas given off or fizzes or bubbling or it bubbles or fizzing and bubbles or effervescence or carbon dioxide is produced or carbon dioxide given off or a new gas is made or it fizzed or fizzed or bubbles were produced or bubbles are produced or gas bubbles or bubbles of carbon dioxide or a gas was produced or a gas was given off or bubbles were given off or fizzing and bubbling
> **Notes / working: Bubbles of gas (fizzing)** — a gas being produced is evidence that a new substance has been made, so a chemical reaction has happened.

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11 (continued) **[1]**

The science club mixes citric acid powder and bicarbonate of soda in a cup of water. They measure the temperature before and after.

Before: 21 °C	After: 13 °C

(c) By how many degrees did the temperature change?

> **Answer:** 8 °C
> **Notes / working:** Change = 21 − 13 = **8 °C** (the temperature went down). A change in temperature is evidence of a chemical reaction.

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12 (continued) **[1]**

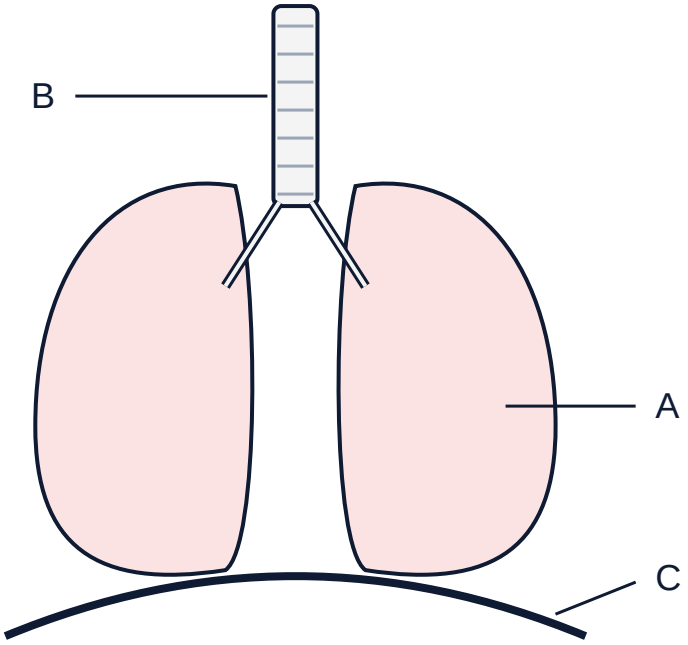
(d) Which observation is **not** evidence of a chemical reaction?

- ☐ A) The colour changes from blue to green.
- ☐ B) Bubbles of a gas are given off.
- ☐ C) The mixture gets hotter.
- ☐ D) Sugar disappears when it is stirred into water.

> **Answer:** D) Sugar disappears when it is stirred into water.
> **Notes / working:** When sugar dissolves no new substance is made — the sugar is still there, spread out in the water (you can taste it, and get it back by evaporating the water). Dissolving is a **reversible physical change**, so D is not evidence of a chemical reaction. Colour change, a gas being produced and a temperature change are the three kinds of evidence.

Paper 2 — 35 minutes — 40 marks

1 Omar's family visits the Red Sea coast. Omar goes snorkelling. He breathes air through a tube called a snorkel. The diagram shows part of his respiratory system. **[1]**



(a) Which letter labels the trachea (windpipe)?

Answer: B
Notes / working: B is the tube that carries air from the throat down towards the lungs — the trachea. A is a lung and C is the diaphragm (the muscle under the lungs).

(b) Name the gas that moves from the air in the lungs into the blood. **[1]**

Answer: oxygen or O2 or O_2 or oxygen gas
Notes / working: Oxygen passes from the air sacs in the lungs into the blood in the capillaries around them.

2 (continued) **[1]**

Omar counts his breaths in one minute.

Activity	Breaths per minute
resting on the beach	16
after swimming hard	34

(c) By how much did his breathing rate increase?

Answer: 18 breaths per minute
Notes / working: 34 − 16 = **18 breaths per minute**.

(d) Explain why Omar breathes faster after swimming hard. **[1]**

Answer: 1 mark: his muscles are working harder so they need more oxygen (accept: to get more oxygen into the blood / to remove more carbon dioxide). Do not accept: 'he needs more air' without oxygen; 'he is tired'.
Notes / working: His muscles are working hard and need **more oxygen** (and make more carbon dioxide). Breathing faster takes more oxygen into the blood and removes the extra carbon dioxide.

3 (continued) **[1]**

Omar sees a dolphin come up to the surface of the sea to breathe. A dolphin is a mammal.

(e) Which statement is correct?

- ☐ A) Dolphins take oxygen from the water using gills, like fish.
- ☐ B) Dolphins have lungs and breathe air, like humans.
- ☐ C) Dolphins do not need oxygen because they live in water.
- ☐ D) Dolphins take in oxygen through their skin only.

Answer: B) Dolphins have lungs and breathe air, like humans.
Notes / working: Dolphins are mammals (vertebrates) with a respiratory system like ours: they have **lungs** and must come to the surface to breathe **air**. Fish are the vertebrates that use gills. Answer B.

4 This question is about growing up. **[1]**

(a) Name the organs in the female reproductive system that make egg cells.

Answer: ovaries or ovary or the ovaries or ovarys
Notes / working: The two **ovaries** make (and release) egg cells.

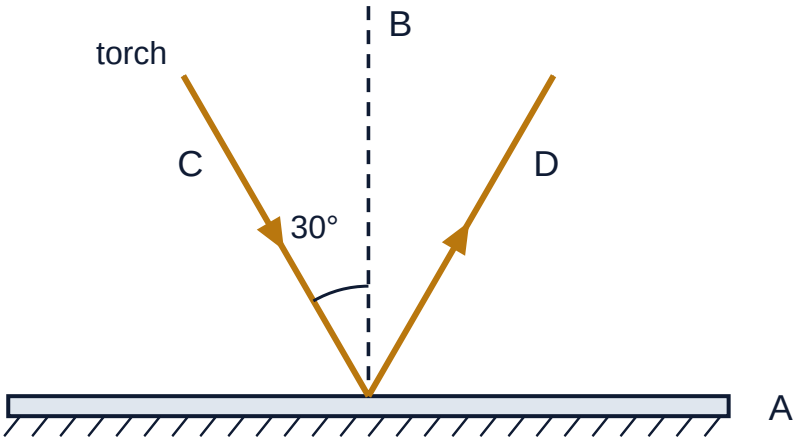
(b) Which statement about puberty is correct? **[1]**

- ☐ A) It starts at exactly the same age for everyone.
- ☐ B) It only happens to boys.
- ☐ C) It is the time when a child's body changes into an adult's body.
- ☐ D) It stops the body from growing taller.

Answer: C) It is the time when a child's body changes into an adult's body.
Notes / working: Puberty is the stage when the body changes from a child's body into an adult's body. It starts at different ages for different people (usually earlier in girls), happens to both boys and girls, and includes a growth spurt. Answer C.

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5 On the boat, Omar shines a torch at a plane mirror. The diagram shows the ray of light. **[1]**



(a) Which letter labels the normal?

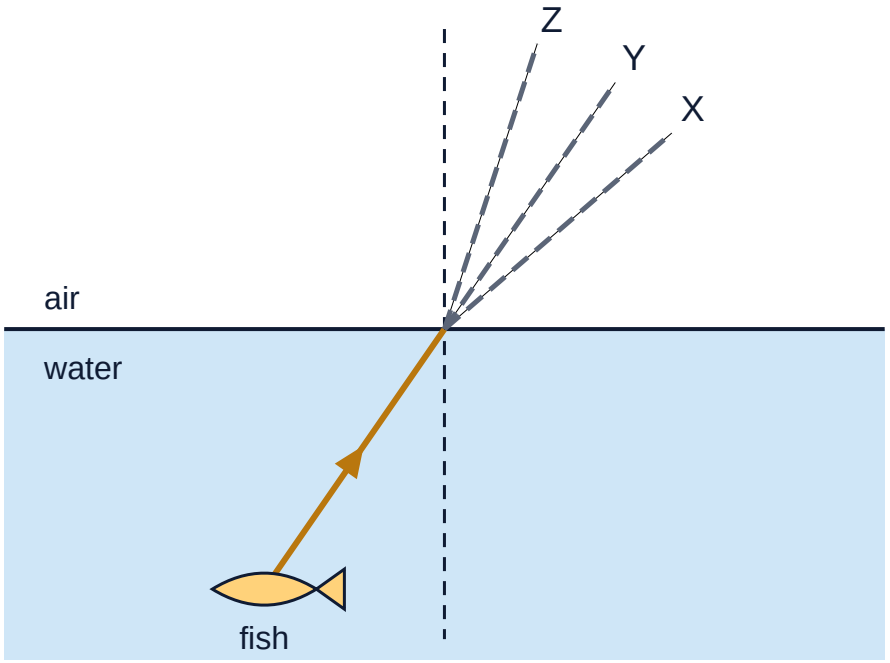
Answer: B
Notes / working: The **normal** is the dashed line drawn at 90° to the mirror where the ray hits it: **B**. A is the mirror, C is the incident ray (from the torch) and D is the reflected ray.

(b) What is the angle of reflection? **[1]**

Answer: 30 °
Notes / working: The angle of incidence (between the incident ray and the normal) is 30°. The angle of reflection is always equal to the angle of incidence, so it is **30°**.
Common mistake: measuring from the mirror and giving 60°.

6 (continued) **[1]**

Omar looks down at a fish from the side of the boat. Light from the fish travels up to the water surface. X, Y and Z show three possible paths for the light after it leaves the water.



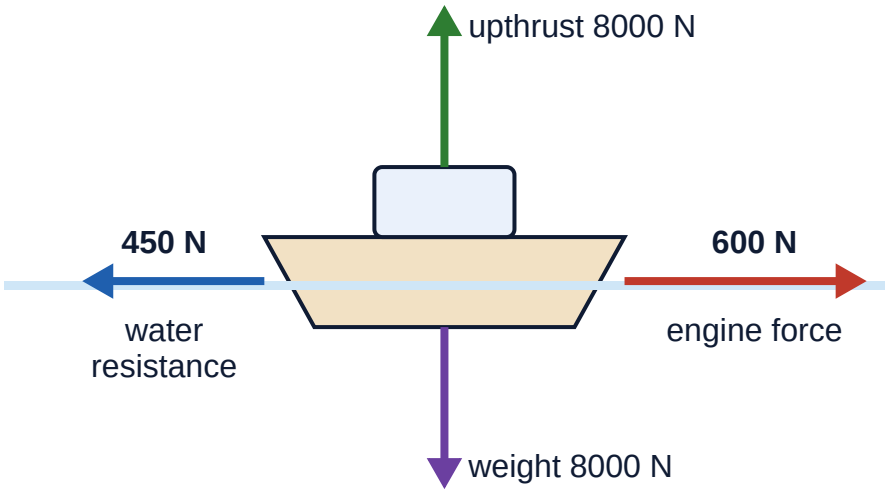
(c) What is the name for the change in direction of light when it passes from water into air?

Answer: refraction or refracting or refracted or refraction of light
Notes / working: Refraction — light changes direction when it passes from one material (medium) into another at an angle.

(d) Which line, X, Y or Z, shows the path of the light in the air? **[1]**

Answer: X or line X
Notes / working: When light passes from water into air at an angle it bends **away from the normal**. Y would be no bending at all (it carries straight on); Z bends towards the normal. So the correct path is **X**.

7 A glass-bottom boat moves across the sea. The diagram shows the forces on the boat. (The up and down arrows are not drawn to the same scale as the side arrows.) **[1]**



(a) Calculate the size of the overall (resultant) force on the boat in the forwards direction.

Answer: 150 N
Notes / working: The side forces act in opposite directions, so subtract: 600 N − 450 N = **150 N** forwards. The upthrust (8000 N) and the weight (8000 N) are equal and opposite, so they cancel.

(b) What happens to the speed of the boat? **[1]**

- ☐ A) It gets faster.
- ☐ B) It gets slower.
- ☐ C) It stays the same.
- ☐ D) The boat stops at once.

Answer: A) It gets faster.
Notes / working: There is an overall force of 150 N **in the direction the boat is moving**, so the boat **speeds up**. Answer A.

(c) Name the force that balances the weight of the boat. **[1]**

Answer: upthrust or up thrust or buoyancy or the upthrust or buoyant force or upthrust force or upward push of water or upward force
Notes / working: The **upthrust** from the water pushes up with 8000 N, the same size as the weight (8000 N), so the boat floats.

(d) The driver switches off the engine. Describe and explain what happens to the motion of the boat. **[1]**

> **Answer:** 1 mark: the boat slows down (and stops) AND because water resistance/friction is now the only side force / acts against the motion with no forward force. Do not accept: 'it stops at once' / 'it sinks'.
> **Notes / working:** The boat **slows down** and eventually stops. With the engine off there is no forward force, so the **water resistance** acting backwards is now an unbalanced force against the motion.

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8 On the beach, Laila puts a plastic bottle with its lid on into the sea water. She adds sand to the bottle, 100 g at a time, and measures how deep the bottle sits in the water. **[1]**

Sand added (g)	How deep the bottle sits (cm)	Floats or sinks?
0	2	floats
100	5	floats
200	8	floats
300	—	sinks

(a) How much deeper does the bottle sit with 200 g of sand than with no sand?

> **Answer:** 6 cm
> **Notes / working:** 8 cm − 2 cm = **6 cm** deeper.

(b) Why does the bottle sink when 300 g of sand is added? **[1]**

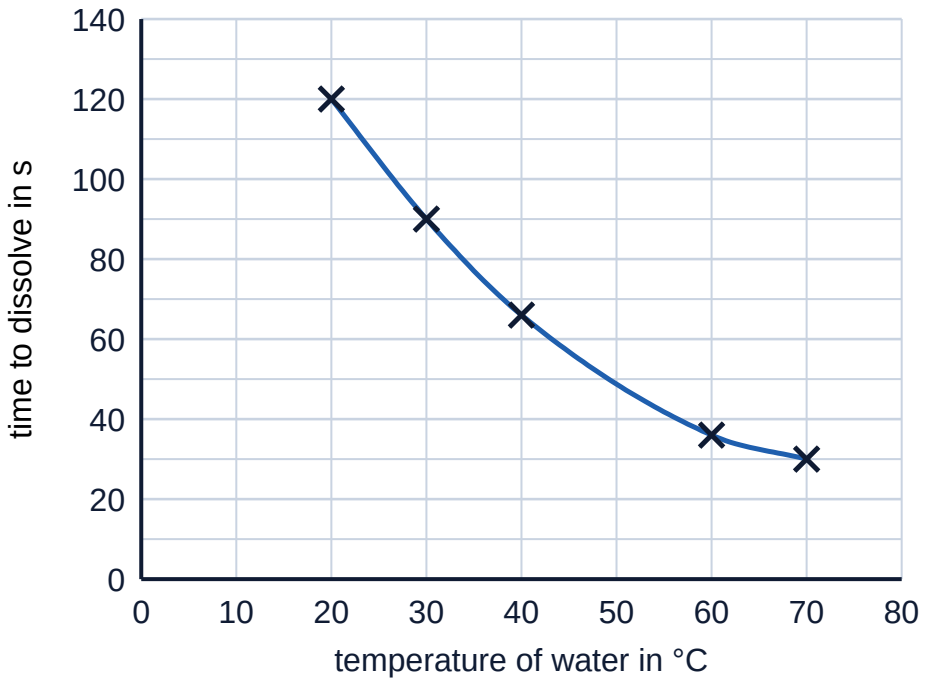
- ☐ A) The bottle changes shape.
- ☐ B) The water pushes up harder than before.
- ☐ C) The sand makes the water less salty.
- ☐ D) Its weight is now greater than the upthrust from the water.

> **Answer:** D) Its weight is now greater than the upthrust from the water.
> **Notes / working:** Adding sand increases the **mass (and so the weight)** of the bottle. At first it floats lower and the upthrust grows to balance it. With 300 g the weight becomes **greater than the biggest upthrust** the water can give, so it sinks. Answer D.

(c) Laila wants to find out more exactly what mass of sand makes the bottle sink. Suggest how she should change her method. **[1]**

> **Answer:** 1 mark: add smaller amounts of sand (e.g. 10 g or 20 g at a time) between 200 g and 300 g / start from 200 g and add sand in small steps. Do not accept: 'repeat the test' alone / 'use a bigger bottle'.
> **Notes / working:** Start with 200 g of sand and **add sand in much smaller steps**, for example 10 g at a time, until the bottle sinks. This finds the sinking mass between 200 g and 300 g more exactly.

9 The science club investigates how the temperature of water affects the time a sugar cube takes to dissolve. Each time they use one sugar cube and 100 ml of water, and they stir at the same speed. They plot their results on a graph. **[1]**



(a) What type of scientific enquiry is this?

- ☐ A) observing over time
- ☐ B) fair testing
- ☐ C) identifying and classifying
- ☐ D) research using books

Answer: B) fair testing
Notes / working: The club **changes one variable** (temperature), **measures one variable** (time) and keeps the others the same (one cube, 100 ml, same stirring). That is a **fair test**. Answer B.

(b) What is the dependent variable in this investigation? (the variable they measure) **[1]**

Answer: 1 mark: the time (taken) for the sugar (cube) to dissolve. Do not accept: 'the sugar' / 'the temperature' / 'how fast it melts'.
Notes / working: The **time taken for the sugar cube to dissolve** — this is what the club measures each time.

(c) Use the graph. How long did the sugar cube take to dissolve in water at 30 °C? **[1]**

Answer: 90 s
Notes / working: Go up from 30 °C on the temperature axis to the plotted cross, then across to the time axis: **90 s**.

(d) The club did not test water at 50 °C. Use the line on the graph to estimate how long the sugar cube would take to dissolve at 50 °C. **[1]**

> **Answer:** 49 s
> **Notes / working:** Go up from 50 °C to the curved line, then across to the time axis. The line is just below the 50 s grid line: about **49 s** (any answer from 46 s to 52 s is accepted).
> Common mistake: drawing a straight line between the points at 40 °C and 60 °C and reading 51 s is close, but always read from the line that is drawn.

(e) Describe the pattern in the results. **[1]**

> **Answer:** 1 mark: the higher the temperature (of the water), the shorter the time to dissolve / the faster the sugar dissolves / ora. Accept: the time goes down as temperature goes up. Do not accept: 'temperature changes the time'.
> **Notes / working: As the temperature of the water increases, the time for the sugar to dissolve decreases** — the sugar dissolves faster in hotter water (the time drops quickly at first, then more slowly).

(f) Use the particle model to explain why sugar dissolves faster in hot water. **[2]**

> **Answer:** 1 mark: (in hot water) the water particles move faster / have more energy. 1 mark: so they hit the sugar particles more often/harder AND the sugar particles break away / spread out between the water particles more quickly. Do not accept: 'the sugar melts' / 'the particles get bigger'.
> **Notes / working:** In hot water the **water particles move faster** (they have more energy). They **hit the sugar particles more often and harder**, so the sugar particles break away from the cube and **spread out between the water particles** more quickly.

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10 Near the Red Sea coast there are salt pans. Sea water is let into wide, shallow pools. Over several weeks the water evaporates in the sunshine and salt is left behind. **[1]**

(a) Explain why getting salt back from sea water shows that dissolving is a reversible change.

> **Answer:** 1 mark: the salt that dissolved is got back (unchanged) / no new substance was made / the salt can be separated out again by evaporating the water.
> **Notes / working:** The salt that dissolved in the sea water is **got back again, unchanged**, when the water evaporates. No new substance was made, so dissolving is a reversible (physical) change.

(b) Which statement describes the evaporation in the salt pans? **[1]**

- ☐ A) It happens only at the surface of the water, below the boiling point.
- ☐ B) It happens only when the water reaches 100 °C.
- ☐ C) It makes large bubbles of gas all through the water.
- ☐ D) It changes the water into salt.

Answer: A) It happens only at the surface of the water, below the boiling point.
Notes / working: Evaporation happens at the **surface** of a liquid at **any temperature below its boiling point** (here, warmed by the Sun). **Boiling** happens at the boiling point, with bubbles of gas forming all through the liquid. Answer A.

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11 (continued) **[1]**

Omar heats 200 g of sea water in a dish until all the water has evaporated. 8 g of salt is left in the dish.

(c) What mass of water evaporated?

Answer: 192 g
Notes / working: Mass of water = mass of sea water − mass of salt = 200 − 8 = **192 g**.

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12 (continued) **[1]**

The table shows the melting points and boiling points of three substances.

Substance	Melting point (°C)	Boiling point (°C)
mercury	−39	357
water	0	100
salt	801	1465

(d) Which substance is a solid at 500 °C?

Answer: salt
Notes / working: At 500 °C: mercury is above its boiling point (357 °C), so it is a gas; water is above 100 °C, so it is a gas; **salt** is below its melting point (801 °C), so it is still a **solid**.

(e) Omar says: "We can use a boiling point to help identify a pure substance." Why is he right? **[1]**

- ☐ A) All liquids boil at 100 °C.
- ☐ B) The boiling point depends on how much liquid there is.
- ☐ C) Each pure substance has its own boiling point.
- ☐ D) The boiling point changes every day.

Answer: C) Each pure substance has its own boiling point.
Notes / working: The temperature at which a substance changes state is a **property of that substance**: each pure substance has its own melting point and boiling point (water 100 °C, mercury 357 °C). It does not depend on the amount. Answer C.

13 Omar's uncle is a diver. He weighs his air tank on a balance. **[1]**

Empty tank: 12.5 kg	Tank filled with air: 15.3 kg

(a) What is the mass of the air in the full tank?

Answer: 2.8 kg
Notes / working: 15.3 kg − 12.5 kg = **2.8 kg** of air.

(b) What does this show about gases? **[1]**

Answer: 1 mark: gases have mass / air has mass (accept: gases have weight). Do not accept: 'gases can be squashed' alone.
Notes / working: The tank got heavier when air was pumped in, so **gases (air) have mass**.

14 (continued) **[1]**
The family keeps their drinks cold in a cool box on the beach.

(c) Which material is best for the walls of the cool box?

- ☐ A) copper
- ☐ B) steel
- ☐ C) plastic foam
- ☐ D) aluminium

Answer: C) plastic foam
Notes / working: The walls must stop heat getting in, so they should be a **poor thermal conductor (a thermal insulator)**. Plastic foam is a thermal insulator; copper, steel and aluminium are metals and are good thermal conductors. Answer C.

15 The mountains near the Red Sea coast contain granite. Granite formed deep underground when magma cooled slowly. Rain, wind and waves break the granite into small grains, which are carried away to the beach and the sea. **[1]**

(a) What type of rock is granite?

Answer: igneous or igneous rock
Notes / working: Granite formed when **magma cooled and solidified**, so it is **igneous** rock (it has large interlocking crystals because it cooled slowly).

(b) What is the name of the process in which the granite is worn away and the grains are carried to the sea? **[1]**

- ☐ A) melting
- ☐ B) erosion
- ☐ C) metamorphism
- ☐ D) solidification

Answer: B) erosion
Notes / working: Erosion — rock is worn away and the pieces are carried away by water, wind or ice. Answer B.

(c) The grains settle on the sea floor. Describe how they could become sedimentary rock, and then later become metamorphic rock. **[2]**

Answer: 1 mark: layers of grains/sediment build up (sedimentation) and are buried, squeezed/compacted (and cemented) into sedimentary rock (e.g. sandstone). 1 mark: the sedimentary rock is buried deeper and is heated and squeezed (high temperature and pressure) WITHOUT melting, which changes it into metamorphic rock. Do not accept: 'it melts' for the second mark.
Notes / working: 1. The grains settle in **layers** (sedimentation). More layers on top **bury** them, and the weight **squeezes** them and sticks them together into **sedimentary rock** such as sandstone.
2. If this rock is **buried deeper**, great **heat and pressure** change it — without melting it — into **metamorphic rock** (sandstone becomes quartzite).

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16 At night on the beach, Omar sees a full moon. **[1]**

(a) Omar sees the full moon on 1 June. About when will he next see a full moon?

- ☐ A) 8 June
- ☐ B) 15 June
- ☐ C) 1 August
- ☐ D) 1 July

Answer: D) 1 July
Notes / working: The Moon takes about **one month (about 29½ days)** to go through all its phases, so the next full moon is about 1 July. (On about 15 June it would be a new moon.) Answer D.

(b) Omar's sister says: "The Moon changes shape during the month because the Earth's shadow covers part of it." Explain the real reason the Moon's shape seems to change. **[1]**

Answer: 1 mark: the Moon orbits the Earth, so we see different amounts of the half of the Moon that is lit by the Sun (accept: we see different parts of its sunlit side as it moves round the Earth). Do not accept: 'the Moon changes shape' / 'clouds cover it'.
Notes / working: The Sun always lights up half of the Moon. As the Moon **orbits the Earth**, we see **different amounts of its sunlit half** from the Earth, so its shape seems to change. The Earth's shadow is not the cause.

(c) Which list shows these planets in order of distance from the Sun, starting with the closest? **[1]**

- ☐ A) Venus, Earth, Mars, Jupiter
- ☐ B) Earth, Venus, Mars, Jupiter
- ☐ C) Mars, Venus, Earth, Jupiter
- ☐ D) Venus, Mars, Earth, Jupiter

Answer: A) Venus, Earth, Mars, Jupiter
Notes / working: The order of the planets from the Sun is Mercury, **Venus, Earth, Mars, Jupiter**, Saturn, Uranus, Neptune. Answer A.